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Lecture 33 4/13/26

Summary of last week:

- Hamilton's principle: A system's time evolution is determined by the EOM's that make S stationary.

where $\mathcal{L} \equiv T - U$.

$S = \int \mathcal{L}\, dt$ is stationary when

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \quad \forall\, q_i \text{ where } q_i \text{ are } \underline{\text{generalized coordinates}}$$

generalized force

generalized momentum

Note: You will often have to find $\vec{v}$ in polar, spherical, etc. coordinates because $T = \frac{1}{2} m \vec{v} \cdot \vec{v}$

BUT you no longer need to use geometry to find $\vec{a}$ (much simpler)

Generalized Coordinates: Minimal # of coordinates that describe positions of all particles in the system given the constraints.

$$\vec{r}_\alpha = \vec{r}_\alpha(q_1, \dots, q_n, t) \qquad \alpha = 1, \dots, N \qquad N = \text{\# of particles}$$

$$q_i = q_i(\vec{r}_1, \dots, \vec{r}_N, t) \qquad i = 1, \dots, n \qquad n \neq N \text{ in general}$$

- Natural coordinates: $\vec{r}_\alpha \longleftrightarrow q_i$ mapping doesn't depend on t
- dof's: # of coordinates that can be independently varied
- holonomic: when # dofs $= n$ (# of q_i's)
- For holonomic systems with generalized coordinates, EOM's are given by:

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \qquad i = 1, \dots, n \qquad \text{(Proof in progress)}$$

- On Friday we practiced $\mathcal{L}$ problems

The $\mathcal{L}$ Approach :

How to solve problems with the $\mathcal{L}$ method:
[Assuming all constraints are holonomic & nonconstraint forces come from a PE function U]

(1) Write down $T, U \Rightarrow \mathcal{L} = T - U$ using any convenient inertial reference frame

(2) Choose a set of generalized coordinates $q_1, \dots, q_n$ & find the coordinates used in (1) in terms of q_i

(3) Rewrite $\mathcal{L}(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$

(4) Write down the n Lagrange equations & simplify. These are EOM's!!

→ solve using dif-eq. methods from the course.

Back to proof from last week:

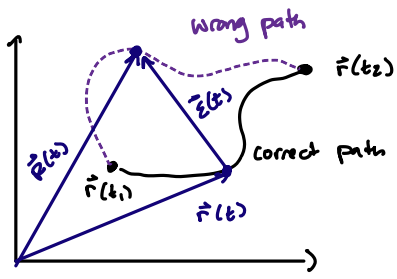
$$\vec{F}_{total} = \underbrace{\vec{F}_{constraint}}_{} + \underbrace{\vec{F}}_{}$$

Constraint forces: tension, Normal force, Static friction

other forces, like gravity, Lorentz, etc.

We want to make the action stationary:

$$S = \int_{t_1}^{t_2} \mathcal{L}\, dt$$



$$\vec{R}(t) = \vec{r}(t) + \vec{\varepsilon}(t) \qquad \vec{\varepsilon} = 0 \text{ at } t = t_1, t_2$$

$$\delta S = S - S_0 = \int_{t_1}^{t_2} \underbrace{\left[\mathcal{L}(\vec{R}, \dot{\vec{R}}, t) - \mathcal{L}(\vec{r}, \dot{\vec{r}}, t)\right]}_{\equiv \delta \mathcal{L}} dt$$

$$\delta T = \tfrac{1}{2} m \dot{\vec{R}}^2 - \tfrac{1}{2} m \dot{\vec{r}}^2 = \tfrac{1}{2} m (\dot{\vec{r}} + \dot{\vec{\varepsilon}})^2 - \tfrac{1}{2} m \dot{\vec{r}}^2 = m \dot{\vec{r}} \cdot \dot{\vec{\varepsilon}} + O(\dot{\varepsilon}^2)$$

$$\delta U = U(\vec{r} + \vec{\varepsilon}, t) - U(\vec{r}, t) = \vec{\varepsilon} \cdot \vec{\nabla} U(\vec{r}) \qquad \text{(from Taylor expansion)}$$

$$\delta \mathcal{L} = m \dot{\vec{r}} \cdot \dot{\vec{\varepsilon}} - \vec{\varepsilon} \cdot \vec{\nabla} U$$

$$\delta S = \int \delta \mathcal{L}\, dt = \int \left(\dot{\vec{\varepsilon}} \cdot m\dot{\vec{r}} - \vec{\varepsilon} \cdot \vec{\nabla} U \right) dt = \int \left(-\vec{\varepsilon} \cdot m\ddot{\vec{r}} - \vec{\varepsilon} \cdot \vec{\nabla} U \right) dt$$

$$= -\int \vec{\varepsilon} \cdot \left[m\ddot{\vec{r}} + \vec{\nabla} U \right] dt$$

$$m\ddot{\vec{r}} = \vec{F}_{total} \Rightarrow \delta S = -\int_{t_1}^{t_2} \vec{\varepsilon} \cdot \left[\vec{F}_{total} + \nabla \vec{U} \right] dt = -\int_{t_1}^{t_2} \vec{\varepsilon} \cdot \left[\vec{F}_{constraint} + \underbrace{\vec{F} + \vec{\nabla} \vec{U}}_{\text{cancel: } \vec{F} = -\vec{\nabla} U} \right] dt$$

$$\delta S = -\int_{t_1}^{t_2} \vec{\varepsilon} \cdot \vec{F}_{constraint}\, dt = 0$$ since all constraint forces are $\perp$ to the surface of movement

($\vec{\varepsilon} \perp$ to all $\hat{q}_i$'s) $\quad \vec{\varepsilon} \perp \vec{F}_{constraint}$

$\Rightarrow \vec{\varepsilon} \cdot \vec{F}_{constraint} = 0$

$\Rightarrow \delta S = 0$ ☑

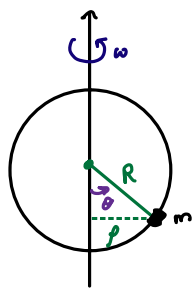
$\Rightarrow$ We showed $\delta S = S - S_0 = 0$ up to $\mathcal{O}(\varepsilon^2)$ when variations occur in the space of minimal generalized coordinates.

$$S = \int_{t_1}^{t_2} \mathcal{L}(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)\, dt$$ being stationary implies:

$$\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \qquad i = 1, \dots, n$$ (by arguments in Ch. 6)

Example $\mathcal{L}$ problem:

<u>Bead on a hoop:</u>



Find EOM's :

$T = \frac{1}{2} m v^2$

$\vec{v} = \frac{d}{dt}(\vec{r})$

Quick reminder of $d\vec{r}$ in various coordinates:

$d\vec{r} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$ Cartesian

$d\vec{r} = d\rho\,\hat{\rho} + \rho\, d\phi\,\hat{\phi} + dz\,\hat{z}$ Cylindrical

$d\vec{r} = dr\,\hat{r} + r\, d\theta\,\hat{\theta} + r\sin\theta\, d\phi\,\hat{\phi}$ Spherical

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{x}\hat{x} + \dot{y}\hat{y} + \dot{z}\hat{z}$$

$$= \dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi} + \dot{z}\hat{z}$$ → let's use cylindrical

$$= \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\hat{\phi}$$

$\vec{v} = \dot{\rho}\hat{\rho} + \rho\dot{\phi}\hat{\phi} + \dot{z}\hat{z}$ Next, choose generalized coordinate θ

$\rho = R\sin\theta$ $\dot{\phi} = \omega$ (given) $z = R - R\cos\theta$

$\dot{\rho} = R\cos\theta\,\dot{\theta}$ $\dot{z} = +R\sin\theta\,\dot{\theta}$

$$\vec{v} = R\cos\theta\,\dot{\theta}\,\hat{\rho} + R\sin\theta\,\omega\,\hat{\phi} + R\sin\theta\,\dot{\theta}\,\hat{z}$$

$$\vec{v}\cdot\vec{v} = \underbrace{R^2\cos^2\theta\,\dot{\theta}^2} + R^2\sin^2\theta\,\omega^2 + \underbrace{R^2\sin^2\theta\,\dot{\theta}^2}$$

simplify to $R^2\dot{\theta}^2$

$$v^2 = R^2\dot{\theta}^2 + R^2\omega^2\sin^2\theta$$

$$T = \tfrac{1}{2}mv^2 = \tfrac{1}{2}mR^2(\dot{\theta}^2 + \omega^2\sin^2\theta)$$

$$U = mgz = mgR(1-\cos\theta)$$

$$\mathcal{L} = T - U = \tfrac{1}{2}mR^2(\dot{\theta}^2 + \omega^2\sin^2\theta) - mgR(1-\cos\theta)$$

$$\frac{\partial\mathcal{L}}{\partial\theta} = \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\theta}}$$

$$\tfrac{1}{2}mR^2\omega^2\,2\sin\theta\cos\theta - mgR\sin\theta = \frac{d}{dt}\left[mgR^2\dot{\theta}\right]$$

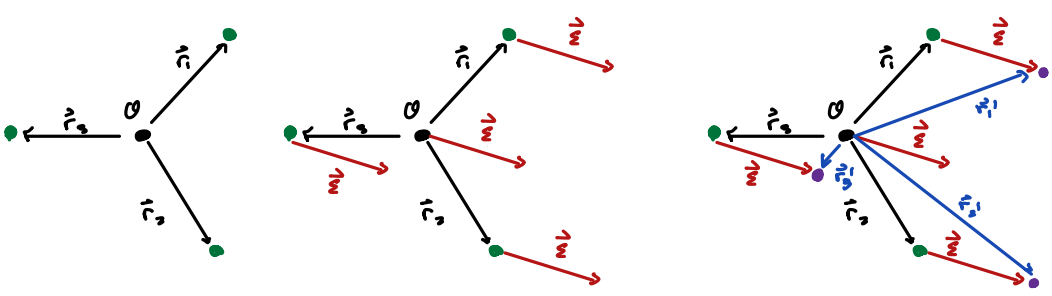
$$mR^2\omega^2\sin\theta\cos\theta - mgR\sin\theta = mR^2\ddot{\theta}$$

$$\boxed{(R\omega^2\cos\theta - g)\sin\theta = R\ddot{\theta}}$$

Conservation Laws & Noether's Theorem

Conservation of momentum $\Longleftrightarrow$ translation invariance

$\vec{r}_1 \to \vec{r}_1 + \vec{s}$, $\vec{r}_2 \to \vec{r}_2 + \vec{s}$, ... , $\vec{r}_N \to \vec{r}_N + \vec{s}$



translation is the same as shifting the position of the origin